

Quiz (Carolina Reaper)

- Q1.** What is the primary purpose of a box-and-whisker plot?
- (A) To display the mean of a dataset
 - (B) To visualize the distribution of data
 - (C) To calculate the standard deviation
 - (D) To identify the mode of a dataset
 - (E) To determine the range of a dataset
- Q2.** A shipping company has the following delivery times (in hours): 2, 4, 6, 8, 10, 12, 14. What is the first quartile (Q_1)?
- (A) 4
 - (B) 6
 - (C) 8
 - (D) 10
 - (E) 12
- Q3.** Which of the following is NOT a characteristic of a box-and-whisker plot?
- (A) Symmetry of the data
 - (B) Skewness of the data
 - (C) Presence of outliers
 - (D) Median of the data
 - (E) Visualization of the raw data
- Q4.** A traffic study recorded the following travel times (in minutes): 15, 20, 25, 30, 35, 40, 45, 50. What is the interquartile range (IQR)?
- (A) 10
 - (B) 15
 - (C) 20
 - (D) 25
 - (E) 30
- Q5.** Which of the following statements is TRUE about outliers in a box-and-whisker plot?
- (A) Outliers are always above the upper whisker
 - (B) Outliers are always below the lower whisker
 - (C) Outliers can be either above the upper whisker or below the lower whisker
 - (D) Outliers are never shown in a box-and-whisker plot
 - (E) Outliers are only shown if they are more than 2 times the IQR away from the median
- Q6.** A logistics company has the following package weights (in pounds): 5, 10, 15, 20, 25, 30, 35. What is the third quartile (Q_3)?
- (A) 15
 - (B) 20
 - (C) 25
 - (D) 30
 - (E) 35
- Q7.** Which of the following types of data is most suitable for a box-and-whisker plot?
- (A) Categorical data
 - (B) Ordinal data
 - (C) Quantitative data
 - (D) Qualitative data
 - (E) Discrete data
- Q8.** A transportation company recorded the following fuel efficiencies (in miles per gallon): 20, 25, 30, 35, 40, 45, 50. What is the median of the data?
- (A) 25
 - (B) 30
 - (C) 35
 - (D) 40
 - (E) 45

Open Ended Questions (FRQ)

- Q9.** A shipping company has the following delivery times (in days): 3, 5, 7, 9, 11, 13, 15. Construct a box-and-whisker plot to display the data and identify any outliers. Show all work and explain your reasoning.
- Q10.** Compare and contrast the use of box-and-whisker plots and histograms for displaying the distribution of quantitative data. Provide an example of a scenario where each type of plot would be more suitable. Show all work and explain your reasoning.

Chef's Tips

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- Tip 1: Ensure students understand the concept of quartiles and interquartile range
- Tip 2: Emphasize the importance of identifying outliers in a dataset
- Tip 3: Encourage students to use real-world examples to illustrate the application of box-and-whisker plots